

COMPSCI 182 Homework 02

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(a)

(1):

$$\text{Let } w^* = X^T a \ (a \in \mathbb{R}^n)$$

$$\therefore X w^* = y$$

$$\therefore X X^T a = y$$

$$\therefore \text{rank}(X) = n$$

$\therefore X X^T$ is invertible matrix

$$\therefore a = (X X^T)^{-1} y$$

$$w^* = X^T (X X^T)^{-1} y$$

$$(2): \therefore w' = w - w^*, \ w_0 = 0$$

$$\therefore w'_0 = 0 - w^* = -X^T (X X^T)^{-1} y$$

(b)

(1): Let $x = U\Sigma V^T$ (U and V are orthogonal matrixs and Σ is $R^{n \times d}$, $\text{rank}(\Sigma) = n$)

We can get $U\Sigma V^T w' = 0$

$$\therefore \Sigma = \begin{pmatrix} \sigma_1 & & 0 & \dots & 0 \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_n & \\ & & & & 0 \dots 0 \end{pmatrix}_{n \times d}$$

$$\therefore U\Sigma = \begin{pmatrix} & & & 0 & \dots & 0 \\ & & & & \ddots & \\ & & & & & \sigma_n \\ & & & & & 0 \dots 0 \end{pmatrix}$$

$$\therefore V^T w' = w''$$

$\therefore$ We can get $[\tilde{X} \ 0_{n \times (d-n)}] w'' = 0$

(2): $w_0' = -x^T a$ ($a \in R^n$)

$$w_0'' = -V^T x^T a$$

$$= -(xV)^T a$$

$$= -(U\Sigma V^T V)^T a$$

$\therefore V$ is orthogonal matrix

$$\therefore w_0'' = -(U\Sigma)^T a$$

$\therefore U$ is orthogonal matrix

$$\therefore U = \begin{pmatrix} | & | & & | \\ e_1 & e_2 & \dots & e_n \\ | & | & & | \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \\ & & & & 0 \dots 0 \end{pmatrix}$$

$$\therefore U\Sigma = \begin{pmatrix} | & | & & | \\ \sigma_1 e_1 & \sigma_2 e_2 & \dots & \sigma_n e_n \\ | & | & & | \end{pmatrix}_{n \times (d-n)}$$

$$\therefore (U\Sigma)^T = \begin{pmatrix} -\sigma_1 e_1 - \\ -\sigma_2 e_2 - \\ \vdots \\ -\sigma_n e_n - \\ 0_{(d-n) \times n} \end{pmatrix}_{d \times n}$$

$$\therefore (U\Sigma)^T a = \begin{pmatrix} -\sigma_1 e_1 a - \\ -\sigma_2 e_2 a - \\ \vdots \\ -\sigma_n e_n a - \\ 0_{(d-n) \times 1} \end{pmatrix}$$

(c)

(1): I have proved in b that $[\tilde{X} \ 0_{n \times (d-n)}] W'' = 0$

$$\begin{aligned}\therefore [\tilde{X} \ 0] W'' &= [\tilde{X} \ 0] \begin{bmatrix} \tilde{w} \\ \tilde{w}_{rest} \end{bmatrix} \\ &= \tilde{X} \tilde{w} + 0 \cdot \tilde{w}_{rest} \\ &= \tilde{X} \tilde{w}\end{aligned}$$

(2): $\because Xw = y, Xw^* = y$

$$\therefore x_i w = y_i, x_i w^* = y_i$$

$$\therefore x_i (w - w^*) = x_i w - y_i = y_i - y_i = 0$$

$$\therefore X = U \Sigma V^T$$

$$\therefore XV = U \Sigma$$

$$\therefore U \Sigma = [\tilde{X} \ 0]$$

$$\therefore XV = [\tilde{X} \ 0]$$

So, the i row of XV is $[\tilde{x}_i^T \ 0]$

$$\begin{aligned}\therefore [\tilde{x}_i^T \ 0] \cdot W'' &= \tilde{x}_i^T \cdot \tilde{w} + 0 \cdot w_{rest} \\ &= \tilde{x}_i^T \tilde{w}\end{aligned}$$

(d)

$$\frac{\partial L_i(w)}{\partial w} = 2 \cdot (y_i - x_i^T w) \cdot x_i \\ = -2 x_i (y_i - x_i^T w)$$

$$\therefore w_{t+1} = w_t + 2\eta x_i (y_i - x_i^T w_t)$$

$$\therefore w'_t = w_t - w^*, x_i^T w^* = y_i$$

$$\therefore y_i - x_i^T w_t = x_i^T (w^* - w_t) = -x_i^T w'_t$$

$$\therefore w_{t+1} = w_t + 2\eta x_i \cdot (-x_i^T w'_t)$$

$$w_{t+1} - w^* = w_t - w^* - 2\eta x_i x_i^T w'_t$$

$$w'_{t+1} = w'_t - 2\eta x_i x_i^T w'_t$$

$$\therefore w'' = V^T w', w' = V w''$$

$$\therefore w''_{t+1} = w''_t - 2\eta V^T x_i x_i^T w'_t$$

$$= w''_t - 2\eta V^T x_i x_i^T V w''_t$$

$$= w''_t - 2\eta (V^T x_i) \cdot (x_i^T V) \cdot w''_t$$

$$\therefore x_i^T V = [\bar{x}_i^T \ 0], V^T x_i = \begin{bmatrix} \tilde{x}_i \\ 0 \end{bmatrix}$$

$\therefore$ If we only look at the first n coordinates, we can get:

$$\tilde{w}_{t+1} = \tilde{w}_t - 2\eta x_i x_i^T \tilde{w}_t$$

(e)

$$\therefore E[L(\tilde{w}_{t+1})] = E[E[L(\tilde{w}_{t+1})|\tilde{w}_t]]$$

$$\therefore E[L(\tilde{w}_{t+1})] \leq (1-p)E[L(\tilde{w}_t)] \Rightarrow E[L(\tilde{w}_t)] \leq (1-p)^T E[L(\tilde{w}_0)]$$

$\therefore \tilde{w}_0$ is a known and definite value

$$\therefore E[L(\tilde{w}_0)] = L(\tilde{w}_0)$$

Substituting $L(\tilde{w}_t)$ into $P(z \geq \epsilon) \leq \frac{E(z)}{\epsilon}$

$$\text{We can get } P(L(\tilde{w}_t) \geq \epsilon) \leq \frac{E[L(\tilde{w}_t)]}{\epsilon}$$

$$\therefore E[L(\tilde{w}_t)] \leq (1-p)^T E[L(\tilde{w}_0)], E[L(\tilde{w}_0)] = L(\tilde{w}_0)$$

$$\therefore P(L(\tilde{w}_t) \geq \epsilon) \leq \frac{L(\tilde{w}_0)}{\epsilon}$$

$$\therefore P(L(\tilde{w}_t) < \epsilon) = 1 - P(L(\tilde{w}_t) \geq \epsilon) \geq 1 - \frac{L(\tilde{w}_0)}{\epsilon}$$

$$\text{Therefore, } P(L(\tilde{w}_t) < \epsilon) \geq 1 - \frac{L(\tilde{w}_0)}{\epsilon}$$

(f)

$$\begin{aligned} L(\tilde{w}) &= (\tilde{x}\tilde{w})^T (\tilde{x}\tilde{w}) \\ &= \|\tilde{x}\tilde{w}\|_2^2 \end{aligned}$$

$$\therefore \|\tilde{x}\tilde{w}\|_2^2 \geq 0, \text{ if and only if } w=0, \|\tilde{x}\tilde{w}\|_2^2 = 0$$

$\therefore L(\tilde{w})$ is non-negative and is only equal to zero at $\tilde{w}=0$

(g)

$$\text{Let } \Delta \tilde{\mathbf{W}}_t = \tilde{\mathbf{W}}_{t+1} - \tilde{\mathbf{W}}_t$$

$$L(\tilde{\mathbf{W}}_t) = \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \tilde{\mathbf{W}}_t$$

$$L(\tilde{\mathbf{W}}_{t+1}) = L(\tilde{\mathbf{W}}_t + \Delta \tilde{\mathbf{W}}_t)$$

$$= (\tilde{\mathbf{W}}_t + \Delta \tilde{\mathbf{W}}_t)^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} (\tilde{\mathbf{W}}_t + \Delta \tilde{\mathbf{W}}_t)$$

$$= \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \tilde{\mathbf{W}}_t + \Delta \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \tilde{\mathbf{W}}_t + \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \Delta \tilde{\mathbf{W}}_t + \Delta \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \Delta \tilde{\mathbf{W}}_t$$

$$\therefore L(\tilde{\mathbf{W}}_{t+1}) - L(\tilde{\mathbf{W}}_t) = \Delta \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \tilde{\mathbf{W}}_t + \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \Delta \tilde{\mathbf{W}}_t + \Delta \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \Delta \tilde{\mathbf{W}}_t$$

$$\therefore A = (\tilde{\mathbf{W}}_{t+1} - \tilde{\mathbf{W}}_t)^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \tilde{\mathbf{W}}_t + \tilde{\mathbf{W}}_t^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} (\tilde{\mathbf{W}}_{t+1} - \tilde{\mathbf{W}}_t)$$

$$B = (\tilde{\mathbf{W}}_{t+1} - \tilde{\mathbf{W}}_t)^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} (\tilde{\mathbf{W}}_{t+1} - \tilde{\mathbf{W}}_t)$$

(h)

I have proved in d that $\tilde{w}_{t+1} - \tilde{w}_t = -2\eta \tilde{x}_i \tilde{x}_i^T \tilde{w}_t$

$$\begin{aligned} A &= (\tilde{w}_{t+1} - \tilde{w}_t) \tilde{x}^T \tilde{x} \tilde{w}_t + \tilde{w}_t^T \tilde{x}^T \tilde{x} (\tilde{w}_{t+1} - \tilde{w}_t) \\ &= -2\eta (\tilde{x}_i \tilde{x}_i^T \tilde{w}_t \tilde{x}^T \tilde{x} \tilde{w}_t + \tilde{w}_t^T \tilde{x}^T \tilde{x} \tilde{x}_i \tilde{x}_i^T \tilde{w}_t) \\ &= -4\eta \tilde{w}_t^T \tilde{x}^T \tilde{x} \tilde{x}_i \tilde{x}_i^T \tilde{w}_t \end{aligned}$$

$$\begin{aligned} E[A|\tilde{w}_t] &= \frac{-4\eta}{n} \sum_{i=1}^n \tilde{w}_t^T \tilde{x}^T \tilde{x} \tilde{x}_i \tilde{x}_i^T \tilde{w}_t \\ &= \frac{-4\eta}{n} \tilde{w}_t^T \tilde{x}^T \tilde{x} \left(\sum_{i=1}^n \tilde{x}_i \tilde{x}_i^T \right) \tilde{w}_t \end{aligned}$$

$$\therefore \tilde{X}^T = [\tilde{x}_1^T, \tilde{x}_2^T, \dots, \tilde{x}_n^T]$$

$$\therefore \tilde{X}^T \tilde{X} = \tilde{x}_1 \tilde{x}_1^T + \dots + \tilde{x}_n \tilde{x}_n^T$$

$$\begin{aligned} \therefore E[A|\tilde{w}_t] &= \frac{-4\eta}{n} \tilde{w}_t^T (\tilde{X}^T \tilde{x})^2 \tilde{w}_t \\ &= \frac{-4\eta}{n} \tilde{w}_t^T \tilde{X}^T \tilde{X} \tilde{X} \tilde{w}_t \\ &= \frac{-4\eta}{n} (\tilde{X}^T \tilde{x} \tilde{w}_t)^T (\tilde{X}^T \tilde{x} \tilde{w}_t) \\ &= \frac{-4\eta}{n} \|\tilde{X}^T \tilde{x} \tilde{w}_t\|^2 \end{aligned}$$

$$\therefore \text{rank}(\tilde{X}) = n$$

$$\therefore \sigma_{\min}(\tilde{X}^T) > 0$$

$$\therefore \|\tilde{X}^T \tilde{x} \tilde{w}_t\|^2 \geq \sigma_{\min}^2 \|\tilde{x} \tilde{w}_t\|^2$$

$$\therefore E[A|\tilde{w}_t] \leq \sigma_{\min}^2 \|\tilde{x} \tilde{w}_t\|^2 \left(\frac{-4\eta}{n} \right)$$

$$\begin{aligned} \therefore L(\tilde{w}) &= \tilde{w}_t^T \tilde{X}^T \tilde{x} \tilde{w}_t \\ &= (\tilde{x} \tilde{w}_t)^T (\tilde{x} \tilde{w}_t) \\ &= \|\tilde{x} \tilde{w}_t\|^2 \end{aligned}$$

$$\therefore E[A|\tilde{w}_t] \leq \frac{-4\eta}{n} \sigma_{\min}^2 \|\tilde{x} \tilde{w}_t\|^2$$

(i)

$$\begin{aligned} B &= (\tilde{W}_{t+1}^T - \tilde{W}_t^T) X^T X (\tilde{W}_{t+1} - \tilde{W}_t) \\ &= 4\eta^2 (\tilde{x}_i \tilde{x}_i^T \tilde{w}_t)^T X^T X (\tilde{x}_i \tilde{x}_i^T \tilde{w}_t) \end{aligned}$$

$\therefore x_i^T \tilde{w}_t$ is a scalar

$$\begin{aligned} \therefore B &= 4\eta^2 (x_i^T w_t)^2 \cdot x_i^T X^T X x_i \\ &= 4\eta^2 (x_i^T w_t)^2 \cdot \|X x_i\|^2 \end{aligned}$$

$$\leq 4\eta^2 (x_i^T w_t)^2 \cdot \sigma_{\max}^2 \|x_i\|^2$$

$$E(B|\tilde{w}_t) \leq \frac{4\eta^2}{n} \sigma_{\max}^2 \cdot \left(\sum_{i=1}^n (x_i^T w_t)^2 \right) \|x_i\|^2$$

Let $\|x_i\|^2 \leq m$ ($m > 0$)

$$\therefore \sum_{i=1}^n (x_i^T w_t)^2 = \|X w_t\|^2$$

$$\therefore E(B|\tilde{w}_t) \leq \frac{4\eta^2}{n} \sigma_{\max}^2 \cdot m \cdot \|X w_t\|^2$$

$$\therefore L(\tilde{w}_t) = \tilde{w}_t^T \tilde{X}^T \tilde{X} \tilde{w}_t = \|X w_t\|^2$$

$$\therefore E(B|\tilde{w}_t) \leq \frac{4\eta^2}{n} \sigma_{\max}^2 \cdot m \cdot L(\tilde{w}_t)$$

Therefore, $E[B|\tilde{w}_t] \leq C_2 \eta^2 L(\tilde{w}_t)$

(j)

$$(i): 1 - C_1\eta + C_2\eta^2 < 1$$

$$\eta < \frac{C_2}{C_1}$$

$$(ii): 1 - C_1\eta + C_2\eta^2 > 0$$

$$\text{if } C_1^2 - 4C_2 < 0$$

$1 - C_1\eta + C_2\eta^2 > 0$ always holds true

Therefore, there exists a η so that $0 < 1 - C_1\eta + C_2\eta^2 < 1$

(k)

The original ridge SGD decreases rapidly at first, but then fluctuates around a nonzero error level. In contrast, the feature-augmented SGD keeps decreasing and approaches zero. Thus, the feature-augmented method continues converging toward the ridge solution, while the original ridge SGD reaches a nonzero error floor.

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(a)

Substituting $x_t = V^T(w_t - w^*)$, $a_t = V^T z_t$, $X = U \Sigma V^T$ into (21):

$$\begin{cases} V^T(w_{t+1} - w^*) = V^T(w_t - w^*) - \eta V^T z_{t+1} \\ V^T z_{t+1} = (1-\beta) V^T z_t + \beta V^T (2X^T x_t - 2X^T y) \end{cases}$$

$$\beta V^T (2X^T x_t - 2X^T y) = 2\beta V^T X^T (x_t - y)$$

$$\because y = Xw^*$$

$$\begin{aligned} \therefore 2\beta V^T X^T (x_t - y) &= 2\beta V^T X^T x_t - 2\beta V^T X^T y \\ &= 2\beta V^T V \Sigma^T U^T U \Sigma V^T (w_t - w^*) \end{aligned}$$

$\because U$ and V are orthogonal matrix

$$\therefore U^T U = V^T V = I$$

$$\therefore V^T(w_t - w^*) = x_t$$

$\therefore$ Original formula is equal to $2\beta \Sigma^T \Sigma x_t$

So we can get:

$$\begin{cases} x_{t+1} = x_t - \eta a_{t+1} \\ a_{t+1} = (1-\beta) \cdot a_t + 2\beta \Sigma^T \Sigma x_t \end{cases}$$

$$\text{Therefore: } \begin{cases} x_{t+1}[i] = x_t[i] - \eta a_{t+1}[i] \\ a_{t+1}[i] = (1-\beta) a_t[i] + 2\beta \sigma_i^2 x_t[i] \end{cases}$$

(b)

$$\text{I have proved in a that } \begin{cases} a_{t+1}[i] = (1-\beta) a_t[i] + 2\beta \theta_i^2 x_t[i] \\ x_{t+1}[i] = -\eta(1-\beta) a_t[i] + (1-2\eta\beta\theta_i^2) x_t[i] \end{cases}$$

$$\therefore \begin{bmatrix} a_{t+1}[i] \\ x_{t+1}[i] \end{bmatrix} = \begin{bmatrix} 1-\beta & 2\beta\theta_i^2 \\ -\eta(1-\beta) & 1-2\eta\beta\theta_i^2 \end{bmatrix} \begin{bmatrix} a_t[i] \\ x_t[i] \end{bmatrix}$$

(c)

$$\lambda_{1,i} = \frac{(2-\beta-2\eta\beta\theta_i^2) + \sqrt{D}}{2}$$

$$\lambda_{2,i} = \frac{(2-\beta-2\eta\beta\theta_i^2) - \sqrt{D}}{2}$$

if $D > 0$, they are distinct and real

if $D = 0$, they are repeated and real

if $D < 0$, they are non-real complex conjugates

(d)

(1): If $D=0$, eigenvalues are repeated

$$\text{That's when } \eta = \frac{-4\beta\theta_i^2 + 8\theta_i^2 \pm \sqrt{16(\beta\theta_i^2 - 2\theta_i^2)^2 - 16\beta^2\theta_i^4}}{8\beta\theta_i^4}$$

$$= \frac{(2-\beta)\theta_i^2 \pm 2\sqrt{1-\beta} \cdot \theta_i^2}{2\beta\theta_i^4}$$

$$= \frac{2-\beta \pm 2\sqrt{1-\beta}}{2\beta\theta_i^2}$$

$$\therefore \eta_1 = \frac{2-\beta-2\sqrt{1-\beta}}{2\beta\theta_i^2}, \quad \eta_2 = \frac{2-\beta+2\sqrt{1-\beta}}{2\beta\theta_i^2}$$

(2): I have proved in (a) that $\lambda = \frac{2-\beta-2\eta\beta\theta_i^2 \pm \sqrt{D}}{2}$

$$\because D=0$$

$$\therefore \lambda_1 = \frac{2-\beta-2\beta\theta_i^2 \cdot \eta}{2-\beta-2\beta\theta_i^2 \cdot \frac{2-\beta-2\sqrt{1-\beta}}{2\beta\theta_i^2}}$$

$$= \frac{2-\beta-2\beta\theta_i^2 \cdot \eta}{2}$$

$$= -\sqrt{1-\beta}$$

Similarly, $\lambda_2 = \sqrt{1-\beta}$

$$\because 0 < \beta < 1$$

$$\therefore 0 < 1-\beta < 1$$

$$\therefore |-\sqrt{1-\beta}| \in (0, 1), \quad |\sqrt{1-\beta}| \in (0, 1)$$

Therefore, they are stable

$$(3): \eta_{\max} = \eta_2 = \frac{2-\beta+2\sqrt{1-\beta}}{2\beta\theta_i^2}$$

(e)

$$\therefore \lambda_1, \lambda_2 = 1 - \beta \in (0, 1)$$

$\therefore \lambda_1$ and λ_2 are either greater than 0 or less than 0 at the same time

$$f(0) = 2\eta\beta\theta_i^2$$

$$\therefore \eta > 0, \beta > 0, \theta_i^2 > 0$$

$$\therefore f(0) > 0$$

$$f(1) = 2(2 - \beta - \eta\beta\theta_i^2)$$

If $f(1) > 0$, we can get $-1 < \lambda_1 < 1, -1 < \lambda_2 < 1$

$$2 - \beta - \eta\beta\theta_i^2 > 0$$

$$\eta < \frac{2 - \beta}{\beta\theta_i^2}$$

$\therefore$ There are two distinct eigenvalues

$$\therefore D > 0$$

$$\text{That when } \eta < \frac{2 - \beta - 2\sqrt{1 - \beta}}{2\beta\theta_i^2} \text{ or } \eta > \frac{2 - \beta + 2\sqrt{1 - \beta}}{2\beta\theta_i^2}$$

$$\text{Therefore: } 0 < \eta < \frac{2 - \beta - 2\sqrt{1 - \beta}}{2\beta\theta_i^2} \text{ or } \frac{2 - \beta + 2\sqrt{1 - \beta}}{2\beta\theta_i^2} < \eta < \frac{2 - \beta}{\beta\theta_i^2}$$

(g)

For $\sigma_{\min} = 1$:

$$\eta_{\min} = \frac{2 - \frac{3}{4} - 2\sqrt{1 - \frac{3}{4}}}{2 \cdot \frac{3}{4} \cdot (1)^2} = \frac{1}{6}$$

$$\eta_{\max} = \frac{2 - \frac{3}{4} + 2\sqrt{1 - \frac{3}{4}}}{2 \cdot \frac{3}{4} \cdot (1)^2} = \frac{3}{2}$$

For $\sigma_{\max} = 1$:

$$\eta_- = \frac{2 - \frac{3}{4} - 2\sqrt{1 - \frac{3}{4}}}{2 \cdot \frac{3}{4} \cdot (2)^2} = \frac{1}{24}$$

$$\eta_+ = \frac{2 - \frac{3}{4} + 2\sqrt{1 - \frac{3}{4}}}{2 \cdot \frac{3}{4} \cdot (2)^2} = \frac{3}{8}$$

$$\left[\frac{1}{6}, \frac{3}{2}\right] \cap \left[\frac{1}{24}, \frac{3}{8}\right] = \left[\frac{1}{6}, \frac{3}{8}\right]$$

So we choose $\eta = \frac{1}{3}$

For momentum: $r \geq \sqrt{1 - \beta} = \frac{1}{2}$

$\therefore \frac{1}{2}$ is the best

For ordinary gradient descent:

$$r = \frac{\sigma_{\max}^2 / \sigma_{\min}^2 - 1}{\sigma_{\max}^2 / \sigma_{\min}^2 + 1} = \frac{4 - 1}{4 + 1} = \frac{3}{5}$$

$$\therefore \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\therefore \tau_{\text{momentum}} = 2$$

$$\therefore \left(\frac{3}{5}\right)^2 = \frac{9}{25} > \frac{1}{4}, \left(\frac{3}{5}\right)^3 = \frac{27}{125} < \frac{1}{4}$$

$$\therefore \tau_{\text{ordinary}} = 3$$

(h)

Thus, directions with larger singular values are updated more aggressively, while directions with smaller singular values change more slowly.

(i)

Gradient descent with momentum converges faster overall because momentum preserves part of the previous update direction, allowing the parameters to keep moving in a consistent direction instead of relying only on the current gradient.

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(a)

$$A: \frac{\beta_1 m_{t-1} + (1-\beta_1) g_t}{1-\beta_1^t}$$

$$B: \frac{\beta_2 v_{t-1} + (1-\beta_2) g_t^2}{1-\beta_2^t}$$

(b)

For L_2 , $f_t^{\text{reg}}(\theta) = f_t'(\theta) + \lambda \theta_t$

$$\begin{aligned}\therefore \theta_{t+1} &= \theta_t - \eta (f_t'(\theta) + \lambda \theta_t) \\ &= (1 - \eta \lambda) \theta_t - \eta f_t'(\theta)\end{aligned}$$

Therefore, when $\eta \lambda = \gamma$, $f_t(\theta)$ is equivalent to regular SGD on the L_2 -regularized loss $f_t^{\text{reg}}(\theta)$

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(a)

$$\begin{aligned} \|\tilde{X}w - y\|^2 &= \|Xw - y + Nw\|^2 \\ &= \|Xw - y\|^2 + 2(Xw - y)^T Nw + \|Nw\|^2 \end{aligned}$$

$$\begin{aligned} E[\|\tilde{X}w - y\|^2] &= E[\|Xw - y\|^2] + 2E[(Xw - y)^T Nw] + E[\|Nw\|^2] \\ &= \|Xw - y\|^2 + 2(Xw - y)^T E[N]w + E\left[\sum_{i=1}^m (N_i^T w)^2\right] \end{aligned}$$

$$\because E[N] = 0$$

$$\begin{aligned} \therefore E[\|\tilde{X}w - y\|^2] &= \|Xw - y\|^2 + E\left[\sum_{i=1}^m (N_i^T w)^2\right] \\ &= \|Xw - y\|^2 + mE[(N_i^T w)^2] \end{aligned}$$

$$\because (N_i^T w) = w^T N_i N_i^T w$$

$$\begin{aligned} \therefore E[(N_i^T w)^2] &= w^T E[N_i N_i^T] w \\ &= w^T E[N_i^2] w \end{aligned}$$

$$\because \text{Var}(N_i) = E[N_i N_i^T] - (E[N_i])^T, \quad E[N_i] = 0$$

$$\therefore E[N_i N_i^T] = \text{Var}(N_i) = \sigma^2 I_n$$

$$\therefore w^T E[N_i N_i^T] w = \sigma^2 w^T w$$

$$\begin{aligned} \therefore E[\|\tilde{X}w - y\|^2] &= \|Xw - y\|^2 + m\sigma^2 w^T w \\ &= m\left(\frac{1}{m}\|Xw - y\|^2 + \sigma^2\|w\|^2\right) \end{aligned}$$

$$\begin{aligned} \text{Therefore, } \arg\min_w E[\|\tilde{X}w - y\|^2] &= \arg\min_w m \cdot \left(\frac{1}{m}\|Xw - y\|^2 + \sigma^2\|w\|^2\right) \\ &= \arg\min_w \frac{1}{m}\|Xw - y\|^2 + \sigma^2\|w\|^2 \end{aligned}$$

(b)

$$L(w_t) = \frac{1}{2}(\hat{x}_t w - y)^2$$

$$\frac{\partial L}{\partial w} = \hat{x}_t(\hat{x}_t w_t - y)$$

$$w_{t+1} = w_t - \eta \cdot \frac{\partial L}{\partial w}$$

$$= w_t - \eta \cdot \hat{x}_t(\hat{x}_t w_t - y)$$

$$= (1 - \eta \hat{x}_t^2) w_t + \eta \hat{x}_t y$$

$$\therefore E[w_{t+1}] = E[(1 - \eta \hat{x}_t^2) w_t] + \eta E[\hat{x}_t] y$$

$$= E[(1 - \eta(x + N_t)^2) w_t] + \eta y E[x + N_t]$$

$\therefore N_t$ and w_t are independent

$$\therefore E[w_{t+1}] = E[1 - \eta(x^2 + 2xN_t + N_t^2)] E[w_t] + \eta y(x + E[N_t])$$

$$= (1 - \eta x^2 - 2\eta x E[N_t] - \eta E[N_t^2]) \cdot E[w_t] + \eta y(x + E[N_t])$$

$$\therefore E[N_t] = 0, E[N_t^2] = 6$$

$$\therefore E[w_{t+1}] = (1 - \eta x^2 - \eta 6) \cdot E[w_t] + \eta xy$$

(c)

$$|1 - \eta(x^2 + b^2)| < 1$$

$$-1 < 1 - \eta(x^2 + b^2) < 1$$

$$0 < \eta(x^2 + b^2) < 2$$

$$0 < \eta < \frac{2}{x^2 + b^2}$$

(d)

Let w_f be the final value!

When $t \rightarrow +\infty$, $E[w_{t+1}] = E[w_t] = w_f$

So, $w_f = (1 - \eta(x^2 + \sigma^2))w_f + \eta xy$

$$w_f = \frac{xy}{x^2 + \sigma^2}$$

Therefore, $t \rightarrow +\infty$, $E[w_t] \rightarrow \frac{xy}{x^2 + \sigma^2}$

If there were not noise:

$$\sigma^2 = 0, E[w_t] = \frac{y}{x}$$

5.

See my answer in 2(h) and 2(i)